

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000848	Version:	6.0, CURRENT
	Status:	Effective	Effective Date:	19-Sep-2011
Title:	SM DG57. EMC considerations.			Page 2 of 6

DG-57 Service Manual

EMC CONSIDERATIONS

Procedure description
In order to meet with current EMC (Electro-Magnetic Compatibility) normative, the equipment builds in several ferrites and grounding clamps. This procedure lists all the critical EMC points. This document refers to Master Connection Diagram.

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PROCEDURE 1: EMC CONSIDERATIONS

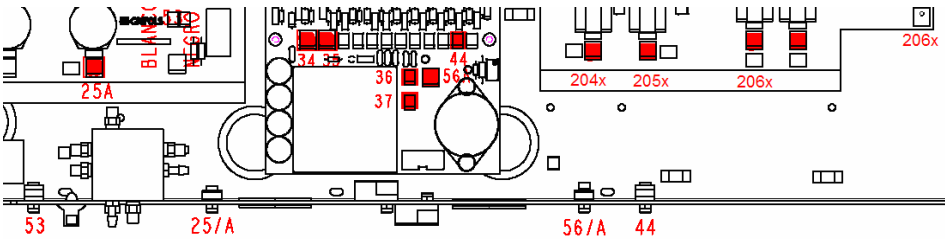
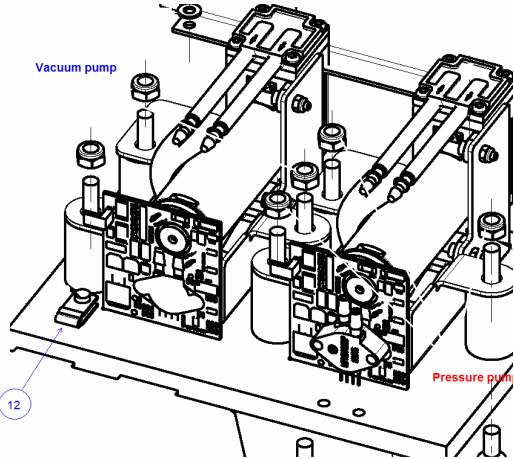
EQUIPMENT

1. ---

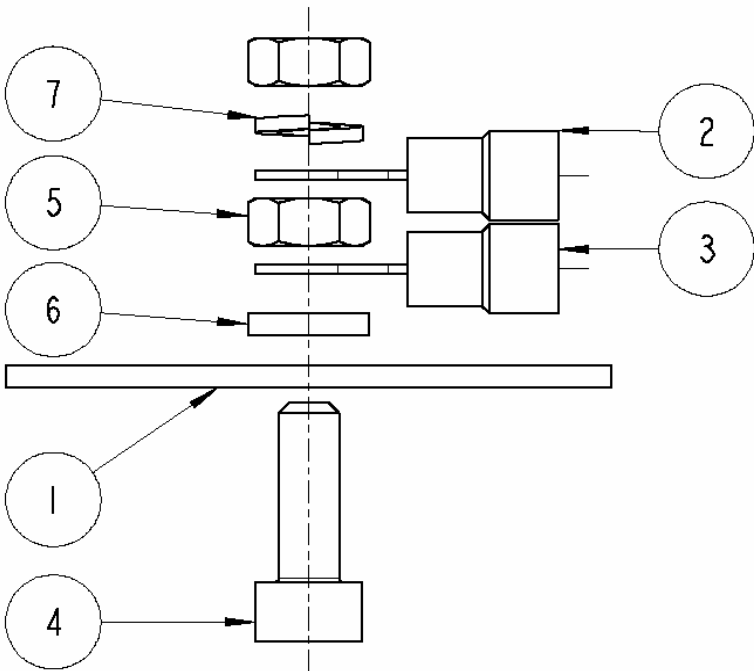
SET UP

1. Turn the instrument off.

PROCEDURE

Process	Expected Results
Shielding and grounding 1. Centrifuge Module: - Cable 25/A centrifuge supply motor is grounded through a clamp. See Picture1. XYZ Module: - Cable 206x supply motor Y is grounded through a clamp. See Picture 1.  Picture 1 2. Pressure-Vacuum Box : - Cable 56/A is grounded through a clamp. See Picture 1. - Cable 56/B is grounded through a clamp inside Pump Box. See Balloon 12 in Picture 2 .  Picture 2 3. Reagents Illumination: Cable 94 Illumination EL screen. - Be sure that the EL cable is covered by a protected cover along the all cable.	--- --- ---

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Process		Expected Results
4.	Main Lock: - Cable 44 main lock is grounded through a clamp. See Picture 1.	---
5.	Illumination status equipment: - Cable 53 is grounded through a clamp. See Picture 1.	---
6.	Main Ground lug. See Ground lug detail in Picture 3. Be sure that the screws are tight.	
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Picture 3		
Balloon Legend:		
1. Chassis 2. Ground lug cable 76 3. Ground lug cable 77		
4. Allen screw 5. Hex nut 6. Tooth lock washer		
7. Split lock washer --- ---		
7.	Check the integrity of the cables 77 and 76. Pull slightly the cable 77 to order to check the union between the cable and the terminal. Repeat the action with the cable 76. See Picture 3	---
8.	Power Supply Grounding. Check that the Negative terminal of Power Supply (see balloon 17 in picture 3) is tight. Check the integrity of the cables.	---

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Process	Expected Results
Picture 4 9. Axis Ground lug. Check the ground cable 82 located in Z tower and 83 located in the Y arm. Both groundings are formed by a ground lug and internal tooth lock washer. 10. Revision of split round ferrites cores with plastic housing There are 7 split ferrite in the following cables: 6, 7, 8, 44 (lock side), 44 (electronic support), 56A (electronic support). These ferrites are drawn as shown picture 4 and are positioned in Master Connection diagram. Picture 4. 11. Revision of split ferrites for flat cable. There is a split ferrite in the cable number 75. This ferrite is positioned in Master Connection diagram. The flat cable wrap three times to the split ferrite. 12. Revision of split square ferrite cores with plastic housing There are 7 split ferrites in the following cables: 56/B (2),25A,45,46, (lock side), 44 (beside connector J13 of PCA IO57-2), 55. These ferrites are drawn as shown picture 5 and are positioned in Master Connection diagram. Picture 5.	--- --- --- ---

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
3.0	2005/05/09	Correction of ground lug cable numbering in step 6, eliminated old file reference.	Document REGD-0000848_v3 replaces document 57080V0M001.
4.0	2010/09/10	- Added information about XYZ module, cable 206x. Changed picture 1. - Changed picture 2, updated to the new vacuum/pressure pumps.	Document updated due to the change of the cable chains and the vacuum/pressure pumps.
5.0	See document header.	Document REGD-0000848_v5 replaces REGD-0000848_v4.	Document version changed.
5.0	See document header.	Date format changed to YYYY/MM/DD inside document change history.	Adoption of ISO 8601 date format to avoid ambiguity.
5.0	See document header.	Document title.	Commercial reference substituted by project reference.
5.0	See document header.	Document header.	Increase information to easy document search.